

Transaction Sequence in Full

Every domain a card authorization touches, in order, with the data crossing each boundary. Traced through a three-arm experiment so the decision and experimentation path is visible rather than assumed.

Version	1.0
Companion to	Flow and data model v1.0
Scenario	US ecommerce credit, CIT, three-arm
Steps	38 across 8 phases

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Participants

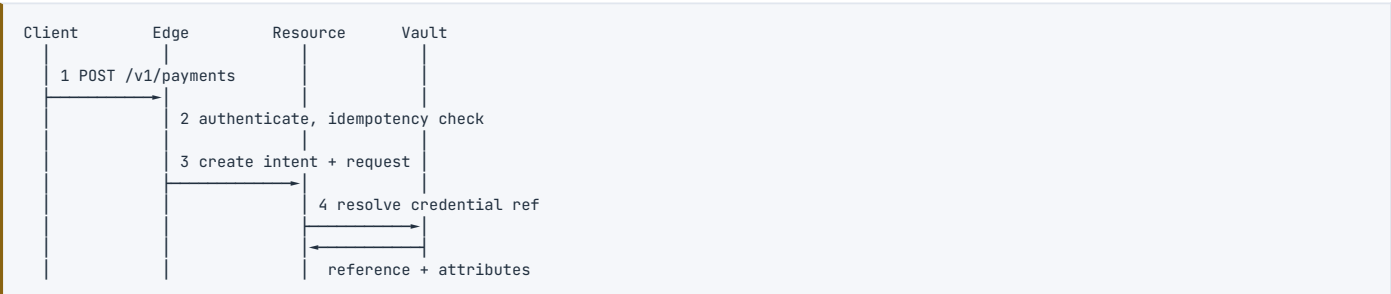
Twenty-one domains touch this transaction. Nine are on the synchronous authorization path and carry a latency budget; the rest are ahead of it, alongside it, or after it.

Domain	Role in this sequence	Path	Budget
Edge	Authn, idempotency, schema, rate	Sync	9 ms
Resource	Obligation, order, intent, request, attempt	Sync	3 ms
Credential Vault	Resolves the credential reference and its enriched attributes	Sync	2 ms
Instrument Intelligence	Supplied the attributes at vault time, not now	Ahead	—
Classification	Resolves the txn type	Sync	1 ms
Screening	Sanctions and watchlist	Sync	12 ms
Control Plane	Constraint set and action space, from a pushed snapshot	Sync	3 ms
Execution Resolver	Txn type to eligible flows	Sync	2 ms
Experimentation	Eligibility, assignment, propensity	Sync	2 ms
Decision Platform	Resolves the strategy bound at each decision point	Sync	per point
Risk	Prediction vector, not a single score	Sync	22 ms
Authentication	Step-up, exemption or none	Sync	3 ms
Message Intelligence	Issuer-specific formatting and enrichment	Sync	3 ms
Routing	Processor, acquirer, network, message model	Sync	4 ms
Processor Adapter	Canonical to network message	Sync	3 ms
Orchestrator	Runs the flow graph, owns deadlines	Sync	6 ms
State Machine	Attempt and intent state transitions	Sync	—
Journal	Immutable event record	Async write	—
Recovery	Only on unknown outcome	Conditional	—
Money Movement	Clearing, settlement, ledger	After	—
Learning	Outcome join and attribution	After	—

The two that are not where you expect

Instrument Intelligence does no work during the transaction — its output arrived with the credential when it was vaulted, which is what removes a remote lookup from the critical path. And the Decision Platform is not a service call in the middle of the flow; it is what resolved the flow's bindings before the transaction started. At runtime each decision point executes an already-bound strategy.

Phase A — ingress and resource



1 · Client to Edge

```
POST /v1/payments
Idempotency-Key: 8f2c...
{
  "amount": 12500, "currency": "USD",
  "payment_method": "pm_1KxA...",           // vault reference
  "capture_method": "automatic",
  "customer": "cus_9Qz...",
  "merchant_id": "mer_4471",
  "channel": "ecommerce",
  "statement_descriptor": "ACME STORE",
  "metadata": {}
}
```

2 · Edge internal

Verifies	API credential · scope · merchant binding idempotency key not previously seen schema · amount bounds · rate limit
Emits	request_id, received_at, client_ip, tls_fingerprint
Never sees	a PAN. The payment_method is already a reference; raw PAN ingress happens only through hosted fields or client-side encryption into the vault (§10 zones)

3 · Edge to Resource

```
CreatePaymentCommand
{
  "request_id": "req_01JX...",
  "idempotency_key": "8f2c...",
  "merchant_id": "mer_4471",
  "amount": 12500, "currency": "USD",
  "instrument_ref": "pm_1KxA...",
  "customer_ref": "cus_9Qz...",
  "channel": "ecommerce", "capture_method": "automatic"
}

Resource creates, implicitly for a standard checkout:
obligation_id  obl_01JX...  customer to merchant, basis=sale
order_id       ord_01JX...  single payer, single instrument
intent_id      pi_01JX...   the object the client holds
request_id     pr_01JX...   operation = AUTHORIZE
attempt_id     att_01JX...  sequence 1
```

4 · Resource to Vault

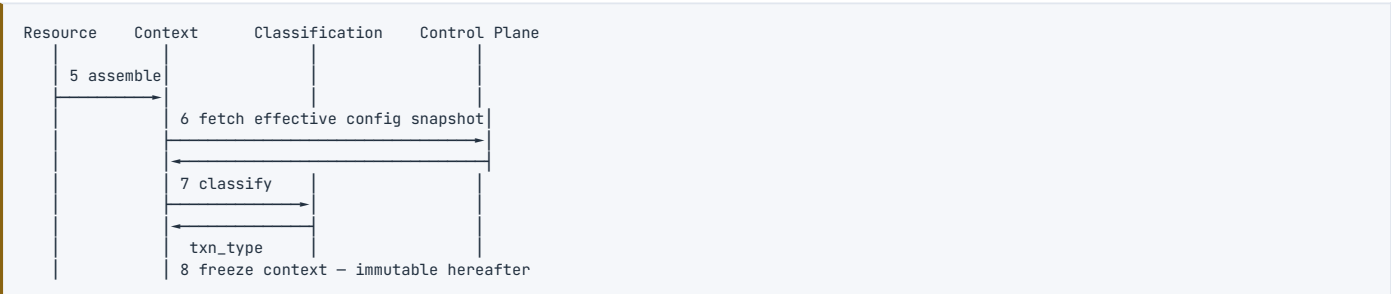
```
Request { instrument_ref, merchant_id, purpose: "AUTHORIZE" }

Response CredentialReference
{
  "credential_id": "cred_7Hs...",
  "type": "NETWORK_TOKEN", "network": "visa",
  "last4": "4242", "expiry": "2029-04",
  "token_requestor_id": "40010030273",
  "network_token_status": "ACTIVE",
  "updater_status": "CURRENT",
```

```
"credential_age_days": 412,  
  
"enriched": {                                // vault-time, §08  
  "issuer_id": "10029",  
  "issuer_country": "US",  
  "product_id": "VISA_SIGNATURE",  
  "funding_type": "CREDIT",  
  "commercial": false, "prepaid": false,  
  "regulated_debit": null,  
  "debit_networks": [],  
  "pinless_eligible": false,  
  "bin_range": "222333",  
  "enrichment_version": "v52",  
  "as_of": "2026-08-30T02:11:00Z",  
  "freshness_class": "SEMI_STABLE"  
}  
}
```

No PAN crosses this boundary. The adapter (step 28) detokenises inside the CDE zone, and the cleartext never returns upward.

Phase B — context assembly



6 · Effective configuration snapshot

Eleven levels of client hierarchy, resolved ahead of time and pushed. Not a graph traversal on the authorization path.

```
EffectiveConfigSnapshot
{
  "snapshot_id": "snp_9931", "version": 4471,
  "hierarchy": {
    "psp_id": "jpmc", "legal_entity": "JPMCB-US",
    "platform_id": null, "payfac_id": null, "isv_id": null,
    "merchant_id": "mer_4471", "submerchant_id": null,
    "location_id": null, "terminal_id": null
  },
  "capability": { "network_tokens": true, "3ds2": true,
    "local_acquiring": ["US"], "rails": ["CARD"] },
  "entitlement": { "authorization_boost": true,
    "autonomy": { "routing": "AUTO",
      "authentication": "ADVISORY" },
    "exploration_budget_bps": 3.0 },
  "preference": { "optimization_posture": "BALANCED",
    "experiment_opt_out": false,
    "default_debit_network": null },
  "pricing_plan": "IC_PLUS_2026A",
  "signature": "..."
}
```

7 · Classification

Input

channel, funding_type, issuer_country, merchant_country, mandate, presence, currency

Output

```
{ "txn_type": "US_ECOM_CREDIT_CIT",
  "classification_version": "v18" }
```

The txn type is what determines which flows are eligible. It is derived from instrument attributes that already arrived with the credential – which is why classification cannot precede vault resolution, and why constraint binding cannot precede classification.

8 · TransactionContext — frozen

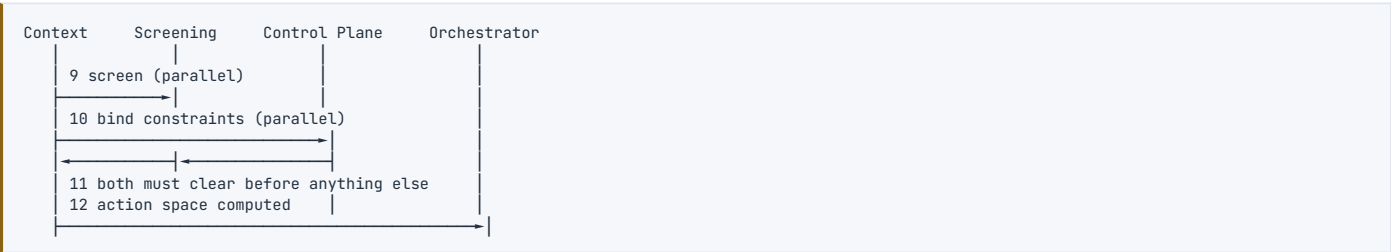
```
TransactionContext          immutable from here
{
  "context_id": "ctx_01JX...",
  "attempt_id": "att_01JX...", "intent_id": "pi_01JX...",
  "txn_type": "US_ECOM_CREDIT_CIT",

  "hierarchy": { ... from snapshot ... },
  "commercial": { "mcc": "5691", "industry": "retail",
    "pricing_plan": "IC_PLUS_2026A" },
  "geography": { "merchant_country": "US",
    "acquiring_country": "US",
    "issuer_country": "US",
    "billing_country": "US", "ip_country": "US",
    "residency_zone": "US" },
  "monetary": { "amount": 12500, "presentment_currency": "USD",
    "settlement_currency": "USD",
    "amount_band": "100-250" },
  "instrument": { ... enriched block from step 4 ... },
```

```
"channel":    { "presence": "CNP", "entry_mode": "ECOM",  
               "wallet": null, "device_id": "dvc_..." },  
"mandate":    { "type": "CIT", "recurring": false,  
               "cof": true, "network_txn_id": null },  
"operation":  { "type": "AUTHORIZE", "capture": "automatic" },  
  
"config_snapshot_ref": "snp_9931@4471",  
"assembled_at": "2026-09-08T14:03:22.104Z"  
}
```

Every downstream domain reads from this object. A module that needs something not in it is either misspecified or reaching outside its contract.

Phase C — screening and constraints



9 · Screening

```
Request { customer_ref, merchant_id, amount, currency,
         issuer_country, ip_country, billing_name_hash }

Response ScreeningResult
{ "decision": "CLEAR", // CLEAR | BLOCK | REVIEW
  "lists_checked": ["OFAC_SDN","EU_CFSP","UK_HMT","INTERNAL"],
  "list_versions": { "OFAC_SDN": "2026-09-07" },
  "duration_ms": 4.2 }
```

BLOCK is terminal. There is no fallback and no retry path – this is the only module in the platform with no degraded mode. A timeout fails the transaction rather than proceeding.

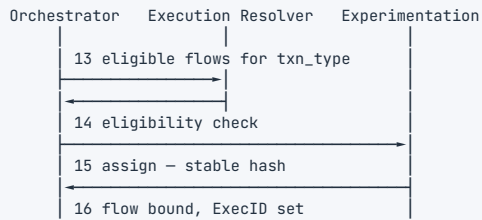
10–12 · Constraint binding and action space

```
ResolvedConstraintSet
{
  "required": [ { "constraint": "REQUIRE_NETWORK_TOKEN",
                  "authority": "MERCHANT", "record": "pol_2201@3" } ],
  "prohibited": [ { "constraint": "NO_CROSS_BORDER_ROUTE",
                    "authority": "PSP", "record": "pol_0044@11" } ],
  "bounded": [ { "constraint": "MAX_RETRY_ATTEMPTS", "value": 3,
                  "authority": "NETWORK", "record": "pol_0912@6" } ],
  "preferred": [ { "constraint": "PREFER_LEAST_COST",
                   "authority": "MERCHANT" } ],
  "sca_applicable": false,
  "bitemporal": { "valid_at": "2026-09-08T14:03:22Z",
                  "known_as_of": "2026-09-08T13:58:00Z" }
}

ActionSpace computed, per decision point
{
  "routing": ["proc_a","proc_b","proc_d"],
  "credential_selection": ["NETWORK_TOKEN"],
  "authentication": ["NONE","3DS_FRICTIONLESS","3DS_CHALLENGE"],
  "message_augmentation": ["STANDARD","DROP_CVV","ADD_DEVICE_DATA"],
  "attempt_strategy": ["STOP","STEP_UP","CRED_SUB","PROC_SUB",
                      "SCHEDULED_RETRY"]
}

proc_c is absent because it would breach NO_CROSS_BORDER_ROUTE.
FPAN is absent because the merchant requires network tokens.
No strategy can select either – not "must not", cannot.
```

Phase D — flow resolution and assignment



13 · Flow resolution

```

Request  { txn_type: "US_ECOM_CREDIT_CIT",
          action_space, config_snapshot_ref }

Response
{
  "baseline_flow": "Flow_1_BAU_v2",
  "eligible_flows": [
    { "flow_id": "Flow_1_BAU_v2",      "role": "BASELINE" },
    { "flow_id": "Flow_2_DropCVV_v1",  "role": "VARIANT",
      "baseline_ref": "Flow_1_BAU_v2", "diff_count": 1,
      "differs_at": "message_augmentation" },
    { "flow_id": "Flow_3_DeviceData_v1", "role": "VARIANT",
      "baseline_ref": "Flow_1_BAU_v2", "diff_count": 1,
      "differs_at": "message_augmentation" }
  ],
  "validated_against_action_space": true
}

```

Both variants differ from baseline at the same decision point, which is why they can share one experiment, one layer and one control arm.

14 · Eligibility

```

ExperimentEligibility
{
  "eligible": true,
  "checks": {
    "merchant_opt_out": false,
    "strategic_account": false,
    "permanent_holdout": false,          // 2% reserved always
    "migration_soak_slice": false,
    "exploration_budget_remaining": true,
    "population_predicate_match": true
  }
}

```

Any false short-circuits to the baseline flow with `eligible=false` and the failing reason recorded. The reason is recorded, not just the outcome – "why was this transaction not in the experiment" is a question that gets asked.

15 · Assignment

```

Request  { experiment_layer: "message_augmentation",
          txn_type, unit_type: "payment_intent",
          unit_id: "pi_01JX...", context_digest }

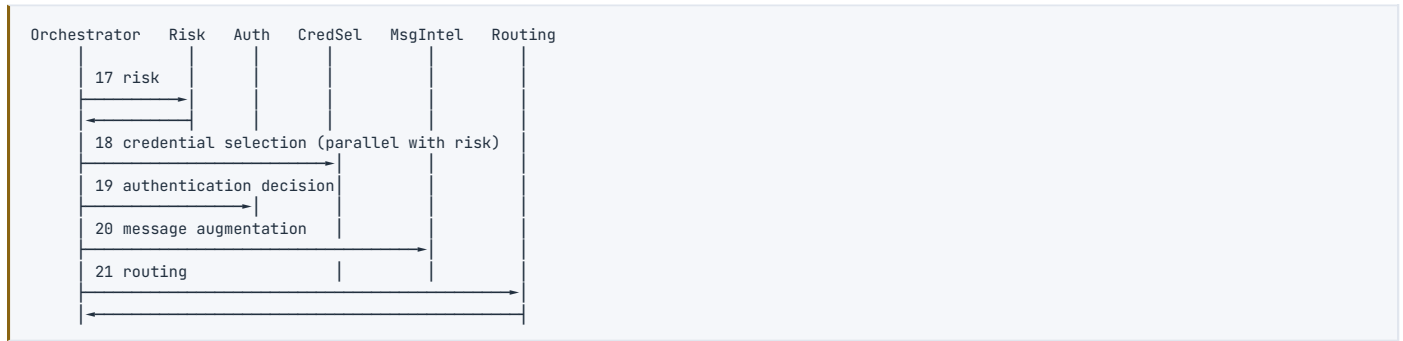
Response Assignment
{
  "experiment_id": "EXP_MsgAug_3Arm",
  "assignment_id": "asg_01JX...",
  "layer": "message_augmentation",
  "salt": "slt_msgaug_2026",
  "unit_type": "payment_intent", "unit_id": "pi_01JX...",
  "bucket": 84,
  "arm": "variant_1",
}

```

```
"flow_id": "Flow_2_DropCVV_v1",  
"propensity": 0.10,  
"allocation": { "control": 0.80,  
                "variant_1": 0.10, "variant_2": 0.10 },  
"assigned_at": "2026-09-08T14:03:22.109Z"  
}
```

bucket = hash(unit_id || salt) mod 100. Deterministic, so a
retry of this intent lands in the same arm. Assignment is
carried on the context, not recomputed.

Phase E — decision execution



Each call carries the same request envelope — the decision contract — and returns the same response shape. That uniformity is what lets a new decision point be added without touching the orchestrator.

Decision request, every point

```

DecisionRequest
{
  "decision_point": "message_augmentation",
  "context_ref": "ctx_01JX...",
  "action_space": ["STANDARD", "DROP_CVV", "ADD_DEVICE_DATA"],
  "bound_strategy": "MsgAug_DropCVV_v1", // from the flow
  "upstream_decisions": { // what already ran
    "risk": { "p_fraud": 0.004, "p_false_decline": 0.031 },
    "credential_selection": { "credential": "NETWORK_TOKEN" }
  },
  "deadline_ms": 3
}

```

Decision response, every point

```

DecisionResponse
{
  "decision_point": "message_augmentation",
  "strategy_id": "MsgAug_DropCVV_v1", "strategy_version": "1.0",
  "action": "DROP_CVV",
  "regime": "POLICY", // POLICY|HEURISTIC|LEARNED|FALLBACK
  "propensity": 1.0, // required even when deterministic
  "considered": ["STANDARD", "DROP_CVV", "ADD_DEVICE_DATA"],
  "rejected": [
    { "action": "STANDARD",
      "reason_code": "NOT_SELECTED_BY_STRATEGY" },
    { "action": "ADD_DEVICE_DATA",
      "reason_code": "NOT_SELECTED_BY_STRATEGY" }
  ],
  "fallback_would_have_been": "STANDARD", // required
  "expected_effect": { "approval_bps": 37 },
  "reason_codes": ["BIN_RANGE_222333_CVV_SENSITIVE"],
  "duration_ms": 1.1
}

```

The two required fields

propensity and **fallback_would_have_been** are mandatory in the response schema. A decision point that cannot populate both fails schema validation and cannot be registered. Without propensity there is no off-policy evaluation; without the fallback there is no attribution. Neither can be backfilled for traffic that has already run.

17 · Risk — a vector, not a score

```

{ "p_fraud": 0.004, "p_dispute": 0.002,
  "p_false_decline": 0.031, "p_abuse": 0.001,
  "p_issuer_decline": 0.118,
  "model_versions": { "fraud": "v19", "false_decline": "v7" },
  "regime": "LEARNED", "propensity": 0.94 }

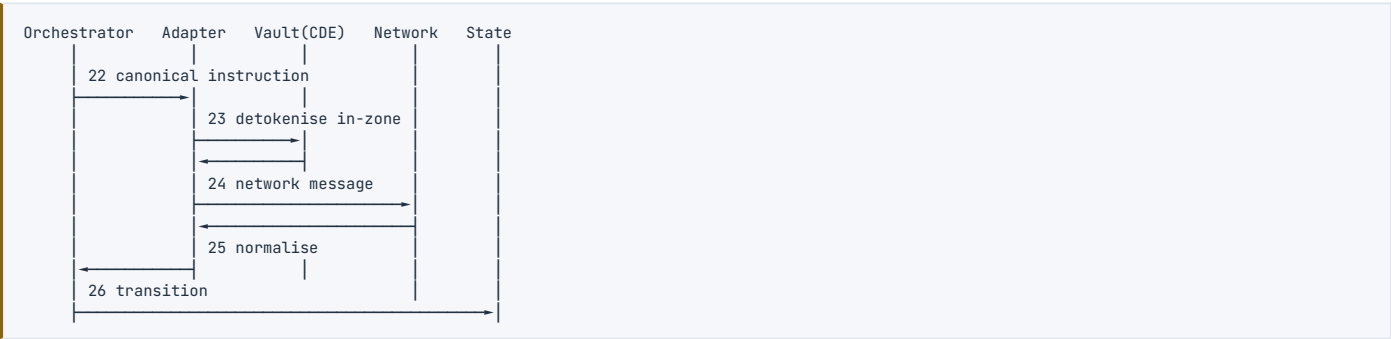
```

Authentication consumes `p_fraud` and `p_false_decline` together – a step-up has a conversion cost, so the tradeoff needs both. Routing consumes `p_issuer_decline`. A single score would force every consumer to treat false declines as free.

21 · Routing output

```
{ "processor": "proc_a", "acquirer": "acq_us_1",
  "network": "visa", "message_model": "DUAL",
  "merchant_account": "ma_4471_us",
  "regime": "LEARNED", "propensity": 0.64,
  "considered": ["proc_a", "proc_b", "proc_d"],
  "rejected": [
    { "action": "proc_b", "reason_code": "HEALTH_DEGRADED" },
    { "action": "proc_d", "reason_code": "HIGHER_EXPECTED_COST" } ],
  "fallback_would_have_been": "proc_b",
  "reason_code": "AUTH_RATE_OPTIMIZATION",
  "expected": { "approval": 0.882, "cost_bps": 174 } }
```

Phase F — authorization



22 · Canonical authorization instruction

```
{
  "attempt_id": "att_01JX...",
  "operation": "AUTHORIZE",
  "amount": 12500, "currency": "USD",
  "credential_ref": "cred_7Hs...", // still a reference
  "credential_form": "NETWORK_TOKEN",
  "processor": "proc_a", "acquirer": "acq_us_1",
  "network": "visa", "message_model": "DUAL",
  "merchant_account": "ma_4471_us",
  "descriptor": "ACME STORE",
  "mandate": { "type": "CIT", "cof": true },
  "authentication": { "performed": false, "sca_applicable": false },
  "message_directives": {
    "omit_fields": ["CVV"], // from step 20
    "include_fields": [],
    "format_profile": "visa_us_issuer_10029_v4"
  }
}
```

23–25 · Adapter, network, normalisation

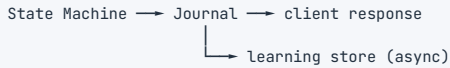
Detokenisation happens inside the CDE zone (§10). The cleartext credential exists only within the adapter process and is never returned upward, logged, or included in any journal event.

Network response, normalised

```
CanonicalResult
{
  "outcome": "APPROVED",
  "code": "APPROVED",
  "class": null,
  "auth_code": "A17R4C",
  "network_txn_id": "0158493201827",
  "avs_result": "Y", "cvv_result": null,
  "retryability": { "retryable": false, "actions_permitted": [] },
  "customer_action": null, "merchant_action": null,
  "raw": { // restricted, §24
    "processor_code": "00", "network_code": "00",
    "iso_field_39": "00"
  },
  "latency_ms": 214
}
```

The raw block is visible to operations and engineering only. Merchants see the canonical code and the actions.

Phase G — record



27 · ExecutionRecord

```

{
  "attempt_id": "att_01JX...", "intent_id": "pi_01JX...",
  "txn_type": "US_ECOM_CREDIT_CIT",
  "exec_id": "Flow_2-DropCVV_v1", "flow_version": "1.0",

  "experiment": {
    "experiment_id": "EXP_MsgAug_3Arm",
    "assignment_id": "asg_01JX...",
    "arm": "variant_1", "bucket": 84, "propensity": 0.10,
    "layer": "message_augmentation"
  },

  "strategies_applied": [
    { "decision_point": "credential_selection",
      "strategy": "CredSel_PreferToken_v4", "regime": "LEARNED",
      "propensity": 0.88 },
    { "decision_point": "authentication",
      "strategy": "Auth_NoSCA_v2", "regime": "POLICY",
      "propensity": 1.0 },
    { "decision_point": "message_augmentation",
      "strategy": "MsgAug_DropCVV_v1", "regime": "POLICY",
      "propensity": 1.0 },
    { "decision_point": "routing",
      "strategy": "Route_LeastCost_v11", "regime": "LEARNED",
      "propensity": 0.64 }
  ],

  "artefact_versions": {
    "config_snapshot": "snp_9931@4471",
    "constraint_set": "v311", "bin_snapshot": "v52",
    "routing_table": "v184", "cost_table": "v77",
    "classification": "v18",
    "models": { "fraud": "v19", "route_auth": "v22" }
  },

  "constraint_provenance": [
    { "constraint": "REQUIRE_NETWORK_TOKEN",
      "authority": "MERCHANT", "record": "pol_2201@3" },
    { "constraint": "NO_CROSS_BORDER_ROUTE",
      "authority": "PSP", "record": "pol_0044@11" }
  ],

  "screening": { "decision": "CLEAR", "list_versions": {...} },
  "result": { "outcome": "APPROVED", "auth_code": "A17R4C" },
  "timings": { "platform_added_ms": 31, "network_ms": 214 }
}

```

28 · Client response — deliberately small

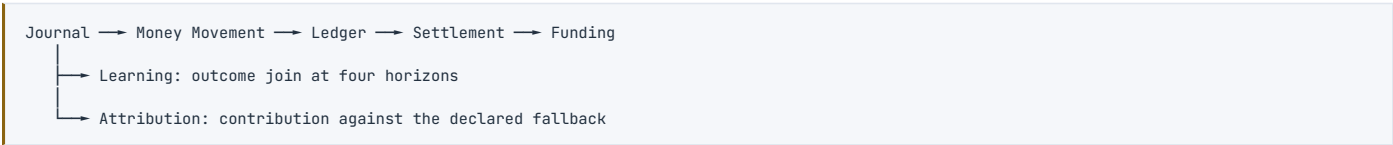
```

{
  "id": "pi_01JX...", "status": "authorized",
  "amount": 12500, "currency": "USD",
  "payment_method": { "type": "card", "network": "visa",
    "last4": "4242", "funding": "credit" },
  "authentication": { "status": "not_required" },
  "authorization": { "status": "approved" },
  "capture": { "status": "pending" },
  "created_at": "...", "updated_at": "..."
}

```

No exec_id, no experiment, no strategies, no processor. Those are available on request at the tiers entitled to them (\$09), not pushed into every response.

Phase H — asynchronous



29–34 · Outcome joins

Horizon	Arrives	Payload added
Authorization	Seconds	approval, auth_code, decline_class
Capture and clearing	Hours to days	captured_amount, interchange_bps, scheme_fees, processor_cost
Fraud	Days to weeks	fraud_flag, fraud_amount, observed_at
Dispute	Weeks to months	dispute_flag, reason_code, liability, outcome

```
OutcomeEvent
{
  "attempt_id": "att_01JX...",
  "horizon": "CLEARING",
  "approval": true,
  "cost": {
    "interchange_bps": 158,
    "scheme_bps": 13,
    "processor_bps": 3,
    "total_bps": 174
  },
  "fraud": {
    "status": "pending",
    "closes_at": "2026-10-08"
  },
  "dispute": {
    "status": "pending",
    "closes_at": "2027-01-06"
  },
  "horizon_complete": ["AUTHORIZATION", "CLEARING"]
}
```

35–38 · Attribution

```
AttributionRecord
{
  "attempt_id": "att_01JX...",
  "experiment_id": "EXP_MsgAug_3Arm",
  "arm": "variant_1",
  "per_strategy": [
    {
      "strategy": "MsgAug_DropCVV_v1",
      "counterfactual": "STANDARD",
      "estimated_contribution_bps": 37,
      "ci": [18, 56],
      "method": "randomised_arm"
    },
    {
      "strategy": "Route_LeastCost_v11",
      "counterfactual": "proc_b",
      "estimated_contribution_bps": 9,
      "ci": [-4, 22],
      "method": "doubly_robust",
      "note": "not randomised in this period"
    }
  ],
  "unattributed_remainder_bps": 6,
  "horizon_status": "PARTIAL - fraud and dispute pending"
}
```

Routing was not randomised here, so its contribution is an off-policy estimate with a wide interval, and the remainder is reported as unattributed rather than apportioned. Only the message augmentation arm carries a randomised estimate.

What each domain may see

Data	Sees it	Never sees it
Cleartext PAN	Adapter, inside CDE only	Every other domain, all logs, the journal
Credential reference	All decision domains	—
Instrument attributes	Classification, constraint, routing, message	—
Customer identifiers	Risk, screening, experimentation (as a hash unit)	Routing, message, adapter
Risk prediction vector	Authentication, routing, attempt strategy	Merchant, except as a band
Experiment assignment	All decision points, journal	Merchant, unless queried
Propensity	Journal, learning	Merchant, integrator
Raw network message	Adapter, operations, engineering	Merchant, platform, PayFac
Constraint provenance	Operations; merchant sees its own authority level only	Other merchants, in every case

Failure paths

Fails at	Disposition	What is recorded
Screening (step 9)	HARD_FAIL — no fallback, no retry	Screening decision, list versions, canonical SCREENING_BLOCKED
Constraint bind (step 10)	HARD_FAIL unless a signed, unexpired last-known-good set is held	Snapshot age and why it was used
Risk (step 17)	FALLBACK per merchant policy: approve, require authentication, or decline	regime: FALLBACK , reason, what the policy chose
Any decision point	FALLBACK to the strategy's declared fallback — never an improvised default	Fallback strategy id, trigger, latency at breach
Message intelligence (step 20)	DEGRADE — base message proceeds	Enrichment absence, so the arm is analysable as degraded
Authorization timeout (step 24)	RECOVER — Recovery Supervisor owns it	Outcome UNKNOWN , recovery case opened, reversal advice issued
Journal write (step 27)	Retried; state rebuilt from events; never blocks the client response	Reconciliation exception if unresolved

The one that changes the experiment

A fallback at the experimented decision point means the transaction did not receive its assigned treatment. It stays in its arm for analysis — dropping it would bias the result toward the cases where the platform happened to be healthy — but it is flagged so the readout can report treatment fidelity alongside the effect. An arm with 4% fallback rate and one with 0.1% are not the same experiment.